Examining the Effects of Exercise and Dark Chocolate Consumption on Blood Endorphin Levels: A Virtual Study Across Age Groups

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1 Abstract

This study investigates the combined effects of exercise and varying cocoa content in chocolate on endorphin levels in women across different age groups. Using a two-way randomized block design, 162 virtual participants from The Islands online resource were divided into six treatment conditions involving combinations of no exercise or light jogging and consuming no chocolate, chocolate with 40% cocoa, or chocolate with 99% cocoa. Blood endorphin levels were measured after the treatments. ANOVA results revealed significant individual effects of both exercise and chocolate on endorphin levels ($p = 0.009$ and $p < 2.44e\text{-}45$, respectively), though no significant interaction between the two factors or age-related differences was observed. Tukey's HSD test further supported the independent influence of exercise and chocolate. While no interaction effect or age-based differences were found, the study reinforces the benefits of exercise and dark chocolate on endorphin levels. Future research should consider additional factors like hormonal or social influences on endorphin fluctuations to better understand their combined effects.

2 Introduction

Endorphins are neurotransmitters that are created inside the brain. They send signals across the nervous system and act as natural painkillers. Endorphins are an essential part of survival because they block pain signals from the brain and help the body cope with any unpleasant feelings. Humans have more than twenty types of endorphins, each having a different role and purpose that can fluctuate throughout the day. Endorphins have several benefits for physical and psychological health, as they reduce depressive symptoms, improve stress and anxiety, and increase self-esteem. Endorphins are released when the body is experiencing pleasurable activities such as eating, exercising, or having sex. In contemporary America, where mental health issues run rampant among the general public and particularly the female population — with more than one in five women living with a mental illness — learning how to manipulate the release of positive endorphins in the brain will directly combat the commonality of depressive symptoms and improve health outcomes overall.

While there are many variables that correlate with endorphin release, previous studies present strong evidence that exercise and dietary choices in particular have a significant positive impact on endorphin levels. According to Medical News Today, intense physical activity releases endorphins that are associated with negative feelings. In contrast, moderate physical activity also releases endorphins, but these are associated with positive feelings such as pleasure. Thus, physical activity, regardless of its intensity, can impact endorphin levels and overall mood. However, we want to specifically look into the types of exercise that increase immediate endorphin production. For example, the concept of "runner's high" is associated with the burst of endorphins experienced when an individual runs, which makes them feel good physically and mentally.

Similarly, the chemical makeup of chocolate directly contributes to an increase in endorphins in the bloodstream. Chocolate is made up of flavonoids, theobromine, and phenethylamine, all of which contribute to mood enhancement. Specifically, the compounds in dark chocolate alter the brain and are important to endorphin release, interacting directly with neurotransmitters such as serotonin and dopamine. These neurotransmitters are essential in regulating moods and can even help prevent memory loss. Therefore, chocolate consumption is essential to implement as a factor in our study.

In addition to our variables of focus, there exist a variety of factors that directly impact endorphin release. For example, females and males experience endorphins differently due to the excess estrogen production in women, which alters hormonal levels and can cause mood swings. Additionally, research indicates that younger individuals have higher endorphin levels than older individuals. Therefore, it is important to consider demographic factors such sex and age when examining endorphin levels.

Due to the researched impacts of both chocolate and exercise on endorphin levels, we've designed a study that will investigate the effects of exercise and chocolate on the release of endorphins. Specifically, due to the interaction that exists between endorphin levels and other factors like age and gender, this study will focus on women of different age groups.

While the impacts of exercise and dark chocolate on endorphin levels have been studied separately, there is

still a gap in our understanding of their combined effects across different age groups in women. The primary research question guiding this study is: How do exercise and varying percentages of cocoa in chocolate interact to affect endorphin levels in women of varying ages? Through this study we aim to provide insights on strategies for improving endorphin levels in women through accessible lifestyle changes.

3 Methods and Procedures

3.1 Participants

The virtual participants utilized in the study originated from the online simulated data-collection resource *The Islands* (https://islands.smp.uq.edu.au/index.php). For reasons that were mentioned previously, all the islanders chosen to participate in the study were women.

Additionally, all the participants sampled were at least 18 years old. Because age has also been shown to impact blood endorphin levels, the women sampled were selected through their classification as one of three different possible age groups we established. The first category was "Young Adults," which grouped ages 18 to 34; the second category was "Middle Aged", which grouped ages 35 to 54; the third category was "Older Adult", which grouped ages 55 and older.

3.2 Determination of Sample Size

The effect size is 0.25, a moderate proportion that represents the factors' likely practical and noticeable effects on blood endorphin levels. We chose to use the standard significance level of $\alpha = 0.05$, which indicates the probability of incorrectly rejecting the null and concluding that there is an effect. We use a power of $1 - \beta = 0.8$ representing the probability of correctly identifying a statistically significant deviation from the null hypothesis, which is that there are no individual effects of exercise and chocolate on blood endorphin levels and no interaction effects between exercise and chocolate on blood endorphin levels. The numerator degrees of freedom is calculated as $(2-1) \times (3-1) = 2$ and the number of groups is calculated as the levels of the product of the levels of the different factors, or $2 \times 3 \times 3 = 18$. Using GPower to compute the minimum sample size necessary to meet our parameters in Figure 1, we see that our sample size needs to be at least 158. In order to create a balanced design, we decided to increase the sample size to 162 so there would be 54 women in each age group and 9 women in each age group in each treatment condition.

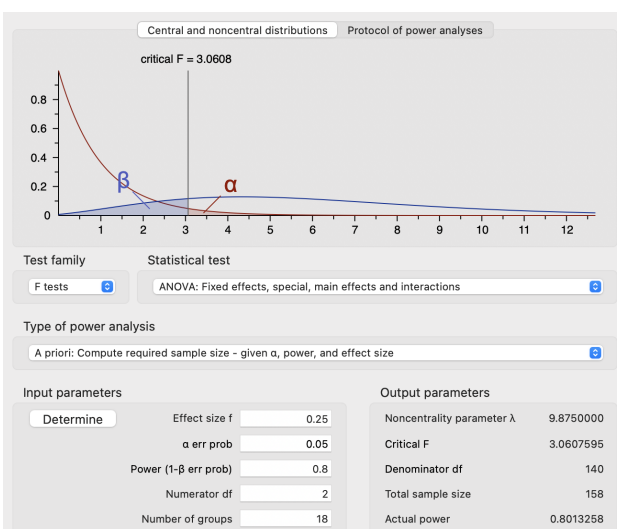

Figure 1: Determining the Required Sample Size Using GPower

3.3 Design

Our focus on the individual effects of exercise and chocolate on overall blood endorphin levels in pg/mL also includes an investigation into their interaction with one another while accounting for the factors of non-interest, age and gender. Thus, we decided to implement a Two-Way Randomized Block Design for the purposes of our study.

In three groups of 54 participants separated by age classification ("Young Adult", "Middle Aged", "Older Adult"), we manipulated the effect of two variables in order to observe their effect on the blood endorphin level of our participants. The first factor was the amount of exercise they would perform, which had two levels: no exercise and light jogging for five minutes. The second factor was the type of chocolate they would eat, which had three levels: none, 50g of dark chocolate with 40% cocoa, and 50g of dark chocolate with 99% cocoa. The factors and factor diagram with corresponding degrees of freedom for the experimental design are displayed in Figure 2 and Figure 3 below, respectively:

Design Parameters			
Response Variable	Blood Endorphin Level in pg/mL		
Treatment 1: Amount of Exercise	No Exercise	Light Jogging for 5 Minutes	
Treatment 2: Type of Chocolate	No Chocolate	50g of Dark Chocolate with 40% Cocoa	50g of Dark Chocolate with 99% Cocoa
Blocking Factor: Age Group	Young Adult (18-34)	Middle Aged (35-54)	Older Adult (55+)

Figure 2: Experimental Factors with Levels

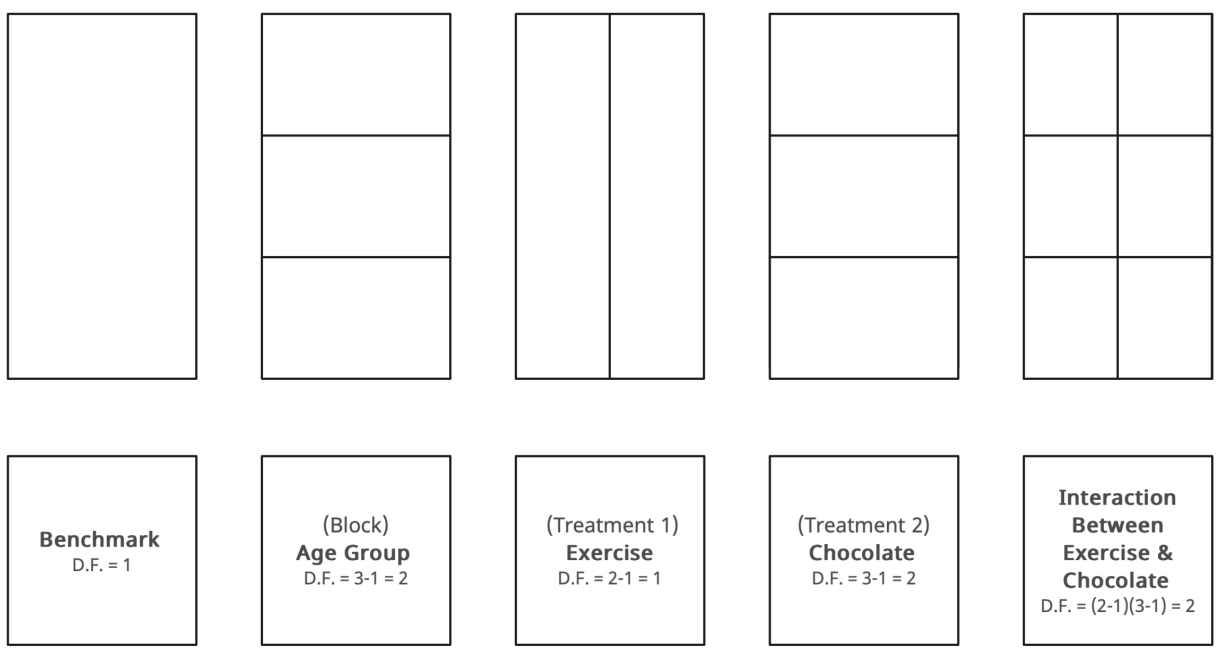

Figure 3: Factor Diagram with Corresponding Degrees of Freedom

3.4 Instruments

Islanders' blood endorphin levels will be collected through the analysis of their blood samples. Islanders in the "Light Jogging" condition will be timed with a timer in order to ensure that their exercise lasts five minutes. After the islanders perform their exercise condition, the participants who are assigned to consume chocolate will be provided with 50g of dark chocolate of the specified cocoa percentage to ingest.

3.5 Procedure

The procedure of data collection implemented systematic assignment methods for convenience purposes and blood measurement directly following the treatment to observe the immediate effect of both exercise and chocolate on endorphin blood level.

We focused on different age groups at a time, so we first sampled 54 young adults, then 54 middle aged adults and, finally, 54 older adults. We visited different islands until we could find a consenting adult fitting the desired age range.

While it was impossible to randomly sample participants from the entire population of islanders due to the large sum of the different islands' occupants and the potential refusal of islanders to participate in the study, we effectuated a systematic method in which to simulate the random assignment of islanders to different treatment conditions so that we could guarantee a balanced design. We numbered the six different treatment conditions: (1) No Exercise and No Chocolate; (2) No Exercise and 40% Cocoa; (3) No Exercise and 99% Cocoa; (4) Light Jogging and No Chocolate; (5) Light Jogging and 40% Cocoa; (6) Light Jogging and 99% Cocoa. Then, we used R to shuffle these numbers in order to create a pseudorandom order of treatment conditions. The result was: (1) No Exercise and No Chocolate; (5) Light Jogging and 40% Cocoa; (3) No Exercise and 99% Cocoa; (6) Light Jogging and 99% Cocoa; (4) Light Jogging and No Chocolate; (2) No Exercise and 40% Cocoa. Finally, when we recorded data in each of the different age blocks, we matched participants to treatment conditions by cycling through the order until all 54 participants were sampled and there were nine observations in each treatment condition in each age group block. For example, the first participant received no exercise and no chocolate, the second participant performed light jogging and ate 40% cocoa chocolate, etc.

We had the islanders execute their assigned exercise treatment first. Immediately after, they were made to ingest the chocolate level they were assigned. Samples of their blood were then drawn and their blood endorphin levels were measured straight after their completion of their assigned exercise and chocolate level. If a participant was assigned to perform no exercise and consume no chocolate, their blood was simply drawn analyzed for their endorphin level.

4 Data Analysis

4.1 Type of Statistical Analysis

In using F-testing within treatments and blocking, we test if the intensity in exercise and levels of cocoa in chocolate impacts a person's endorphin levels. At the same time, we also evaluate if there is a significant difference of endorphin levels within our age block factor and if there is an interaction between our exercise and chocolate variables. We used the baseline of no exercise and chocolate as a control group to compare its average to other treatment groups when it comes to measuring the change in endorphin levels.

4.2 Exploratory Analysis

4.2.1 Boxplots

We perform an exploratory analysis of the data in order to observe its dynamics and gain a general understanding of the its trends before executing a two-way ANOVA.

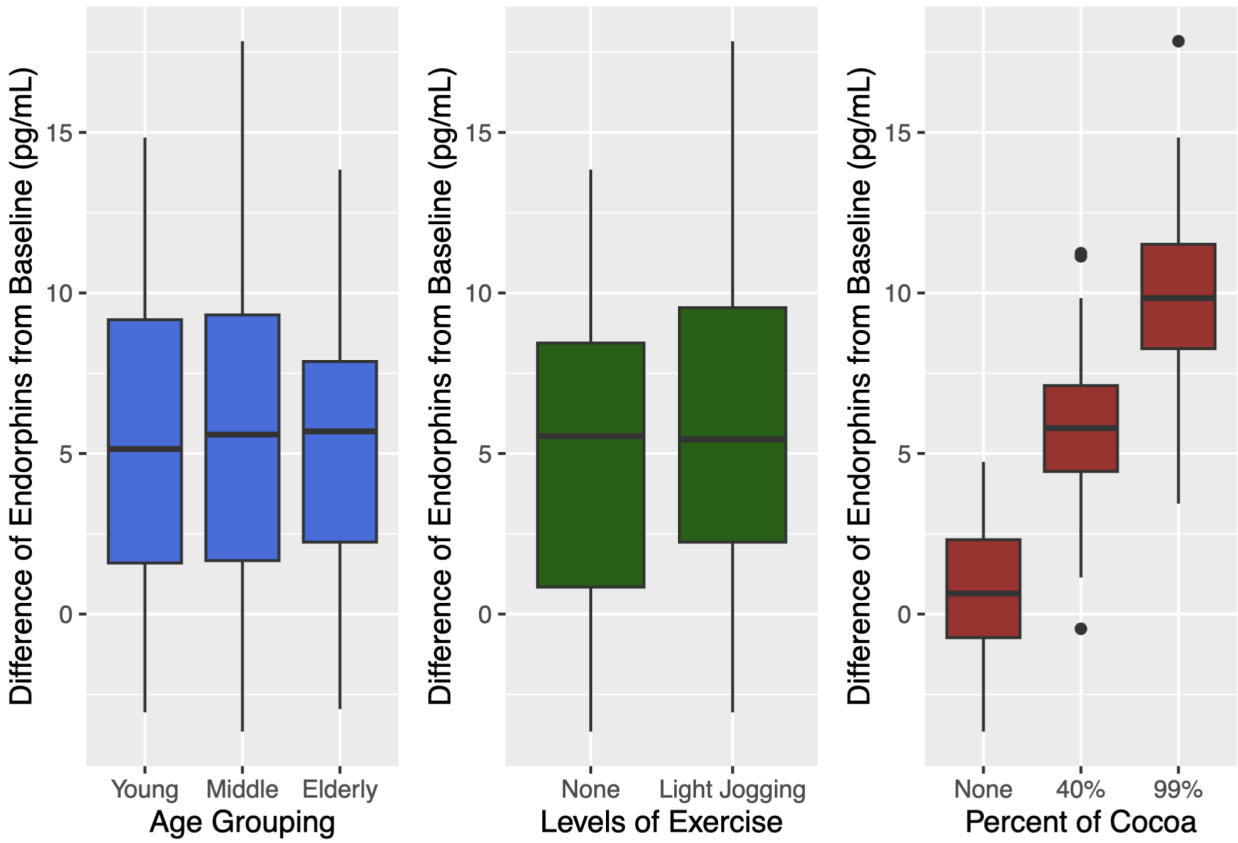

Figure 4: Comparing the Change of Endorphin Levels to Age, Level of Exercise, and Percent of Cocoa

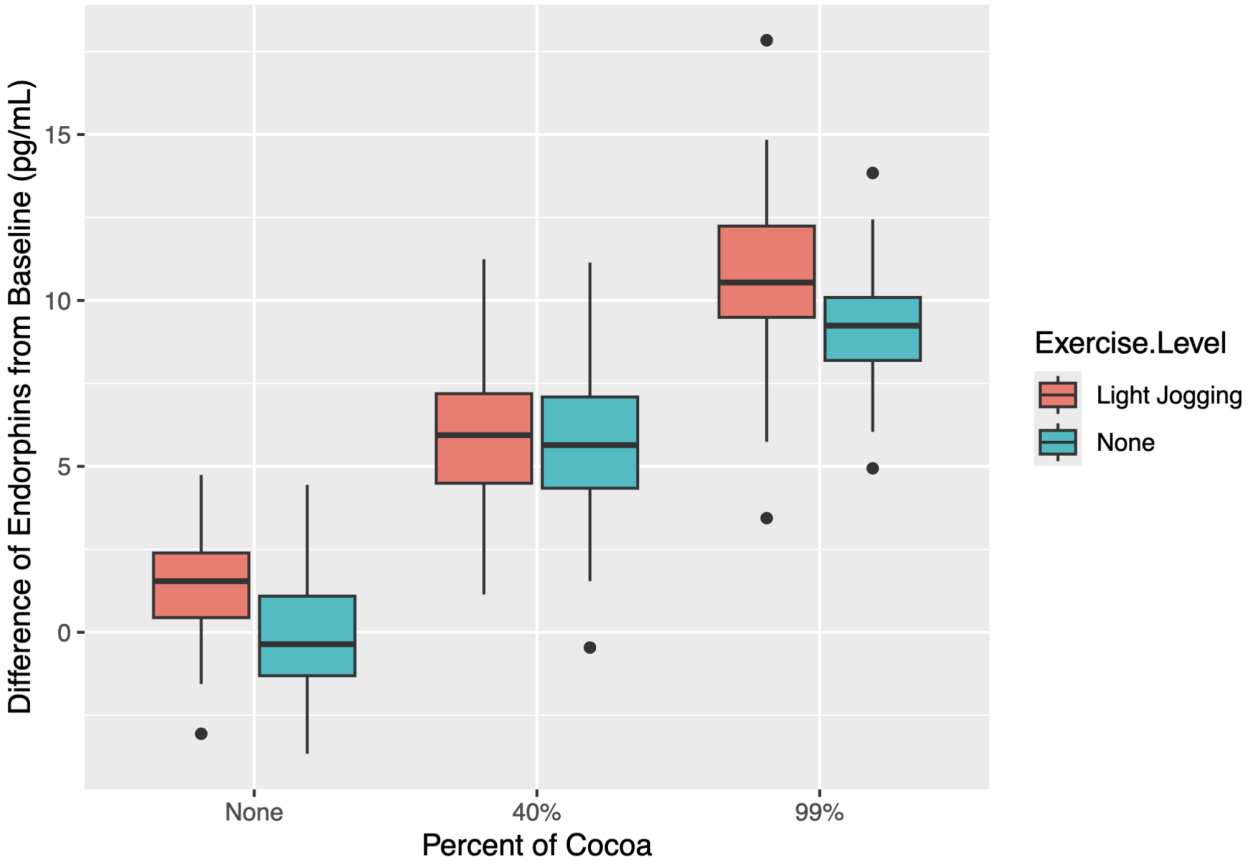

Figure 5: Side-by-Side Boxplot of Exercise Level Grouped by Percent of Cocoa in Chocolate

4.2.2 Interaction Graph

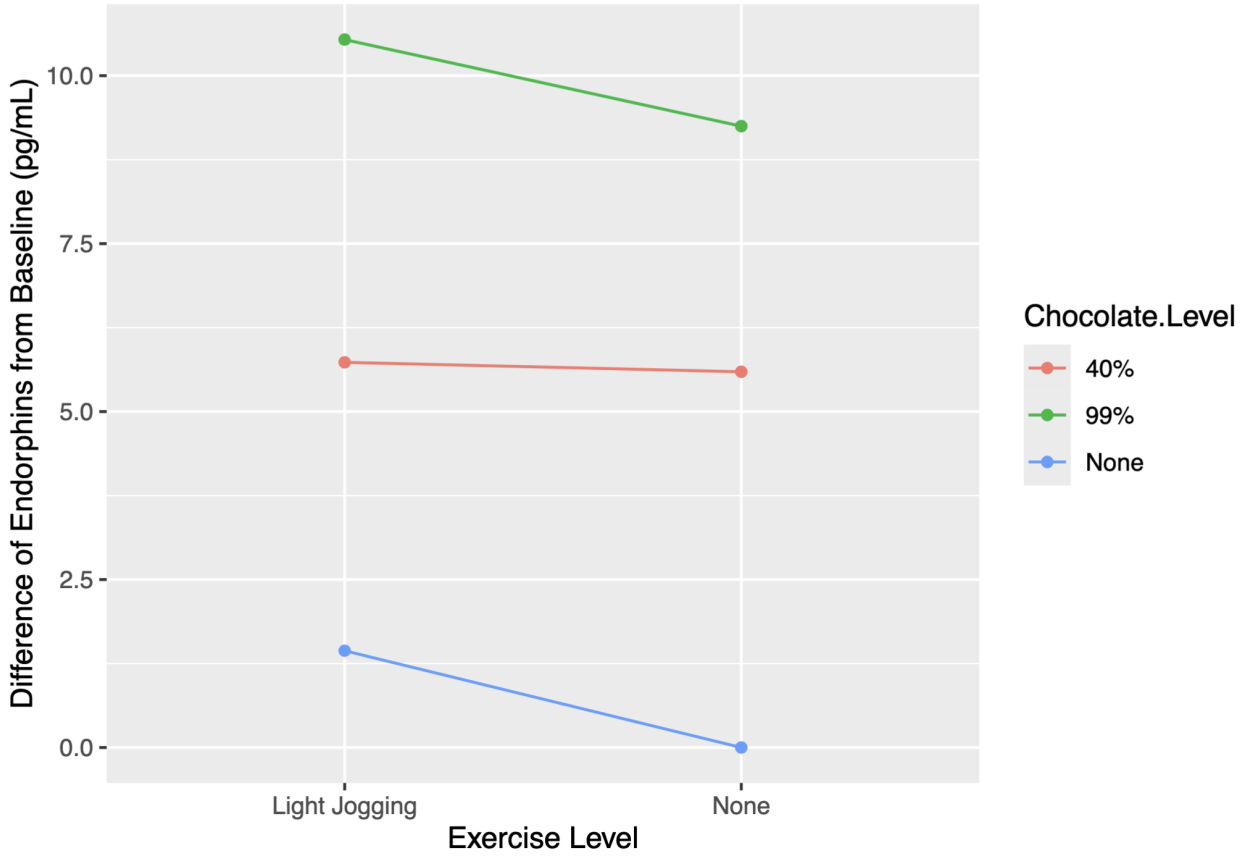

Figure 6: Interaction Plot of Percent of Cocoa and Exercise Level

There is an interaction among chocolate and exercise, which we can derive from the non-parallel lines that are displayed by the interaction graph. However, the lines' slopes are not different from one another by a large amount, indicating that the interaction between the level of chocolate the islander consumers and the level of exercise they perform is small.

5 Results

5.1 ANOVA Analysis

term	df	sumsq	meansq	statistic	p.value
Exercise.Level	1	37.1	37.1	7.02	0.00892
Chocolate.Level	2	2280	1140	215	2.44e-45
Age	2	0.188	0.0938	0.0177	0.982
Exercise.Level:Chocolate.Level	2	13.6	6.82	1.29	0.278
Residuals	154	814	5.28	NA	NA

Figure 7: Two-Way ANOVA Factorial with Blocking Factor

The p-values of 0.00892 and 2.44e-45 for exercise and chocolate respectively means they are statistically significant, while the p-values for age and the interaction between exercise and chocolate are both greater than 0.05, indicating that they are not significant factors. The sums of squares support this, as chocolate and exercise have the largest sums of squares, 2280 and 37.1 respectively, while age and the interaction have sums of squares that are much smaller. This means that blocking by age was not an effective method, as it didn't end up explaining a significant proportion of the total R^2.

5.2 Tukey HSD Adjusted P-Values

	diff	lwr	upr	p adj
None-Light Jogging	-0.9567901	-1.670406	-0.2431742	0.0089231

	diff	lwr	upr	p adj
99%-40%	4.229630	3.182599	5.276660	3.552714e-15
None-40%	-4.942593	-5.989623	-3.895562	0.000000e+00
None-99%	-9.172222	-10.219253	-8.125191	0.000000e+00

Figure 8: Post-Hoc Analyses of Endorphin Levels Between Different Cocoa Levels and Exercise Levels

To prevent the possibility of a Type I error, pairwise comparisons are conducted under this analysis between endorphin levels and between different treatments. Each of the pairwise comparisons displayed above convey significant changes in endorphin levels, as the adjusted p-values are all less than the significance level $\alpha = 0.05$ and all of their corresponding confidence intervals do not contain 0, indicating the small possibility of the differences between the groups being 0.

5.3 Residual Diagnostics

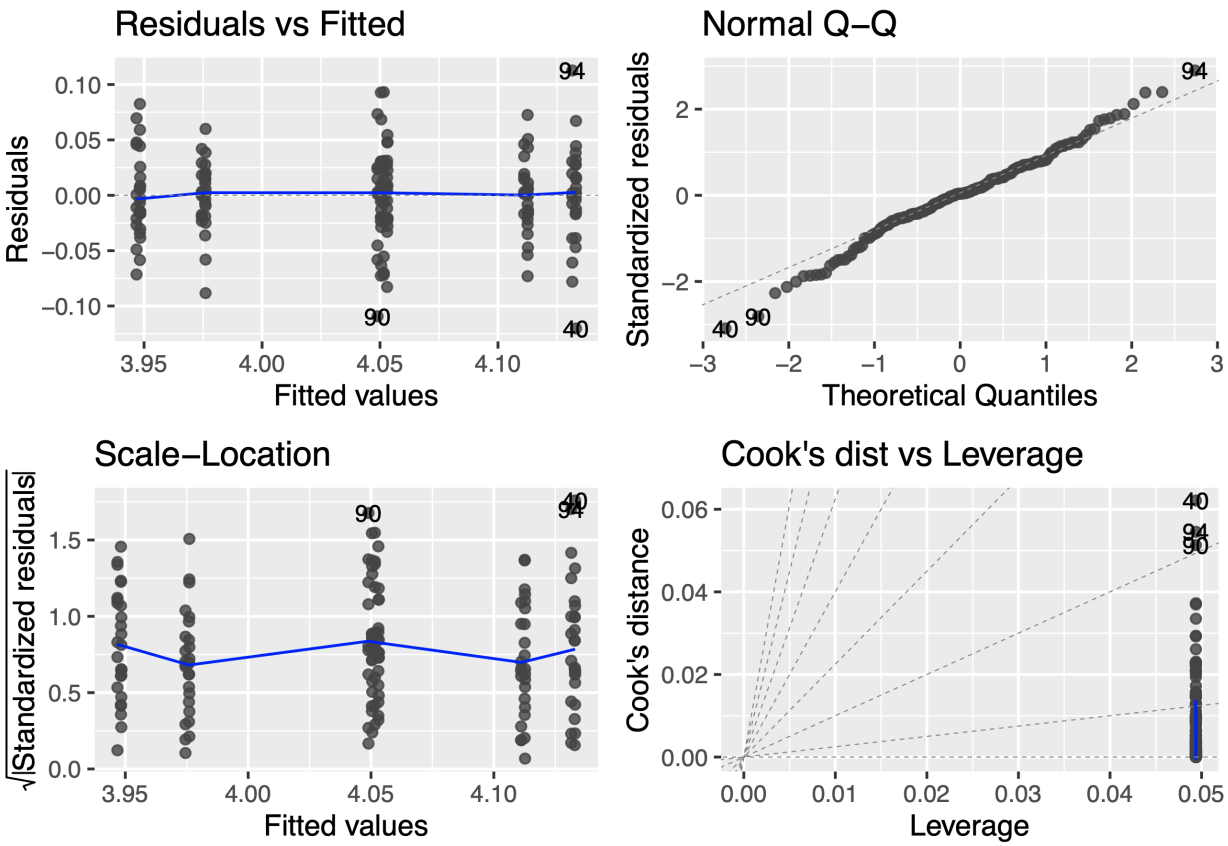

Figure 9: Diagnostic plots Based on ANOVA Model

The residual plots suggest non-constant variance due to points not being randomly scattered around the horizontal reference line in Residuals vs. Fitted and Scale-Location plots. However, the QQ plot does meet the normality assumption, with most of the dots falling near or around the 45-degrees reference line.

6 Discussion

In our initial evaluation of the data, we examined each factor's relationship to endorphin levels through boxplots and an interaction plot. The boxplots in Figure 4 show considerable variability within each age group and exercise level, as evidenced by the wider boxplots and larger vertical spread. However, there were no significant differences in endorphin levels across these factors. Only the percentage of cocoa exhibited a significant impact on endorphin levels. Despite less variability within each level, there were more outliers for cocoa compared to the other factors. Specifically, Figure 5's side-by-side boxplot highlights more outliers as the cocoa level increases. This boxplot, when grouped by exercise level, also suggests slight differences between exercise levels, more so than what is observed in Figure 4.

Figure 6, an interaction plot, further supports the notion that light jogging is associated with a greater change in endorphin levels compared to no exercise. However, our hypothesis regarding the interaction between chocolate and exercise is contradicted by Figure 5. It shows no relation between higher cocoa levels and increased exercise, and the non-parallel lines in Figure 6 suggest no interaction between these factors. Grouping by age also failed to reveal any significant differences in endorphin levels between younger and older participants.

We recognize the limitations of our study, particularly in dividing the age factor into more specific brackets and offering consistent physical activity options. These factors could have helped detect more accurate differences among the treatments and blocks.

After conducting ANOVA analysis, we found significant relationships between both exercise and chocolate levels and endorphin levels, with p-values of approximately 0.009 and 2.44e-45, respectively, as shown in the table in Figure 7. However, the p-values for age and the interaction between chocolate and exercise were much higher than 0.05 (0.98 and 0.28), indicating that neither age grouping nor the interaction between chocolate and exercise was statistically significant. This suggests that while chocolate and exercise individually have strong effects on endorphin levels, their interaction and the age factor do not.

Our Tukey HSD analysis further compared endorphin levels between treatment groups using pairwise comparisons. As seen in the table in Figure 8, there was a statistically significant difference between each level of exercise and chocolate, with p-values below 0.05 and confidence intervals that did not include zero. This confirms that both chocolate and exercise affect endorphin levels in distinct ways.

The residual plots in Figure 9 display values randomly scattered vertically and approximate linearity, which suggests that the assumption of constant variance may not be fully met without transforming the model. However, for the most part, the assumptions of normality and homoscedasticity are met.

Based on the results of the ANOVA and Tukey HSD analyses, we conclude that chocolate (specifically the cocoa content) and physical activity do significantly affect endorphin levels in women. However, we found no evidence of a significant interaction between chocolate and exercise, nor did we observe any effects based on age in this experiment.

We acknowledge several factors that may have influenced our findings and should be considered in future research. The time of day was not accounted for, which could have a controlled effect on endorphin levels, as they fluctuate with daily activities. The variability within each age bracket, as observed in the boxplots, suggests that biological factors (such as hormones and menstrual cycles) or environmental factors (such as work and relationships) may also play a role in affecting women's endorphin levels. Controlling for these factors, either by blocking or holding them constant, would help refine future studies on the relationship between endorphins, chocolate, and exercise.

In summary, our study supports the conclusion that chocolate and exercise influence endorphin levels in women. Promoting these accessible habits could potentially lead to healthier lifestyles.

7 References

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